

Dimension-Free Transformer: The Semi-Tensor Product Approach for Heterogeneous Sequence Modeling

Abstract—While Transformers have revolutionized sequence modeling, their reliance on rigid, fixed-shape tensors imposes strict dimensional constraints and necessitates inefficient zero-padding for variable-length sequences. To address these limitations, we propose the Dimension-Free Transformer (DFT). By replacing standard dot-product attention with the Semi-Tensor Product (STP), DFT decouples the dimension matching requirements of Query/Key/Value ($Q/K/V$) projections. Furthermore, we introduce Projection-Padding, a technique that linearly maps variable sequences to a nominal length, thereby eliminating artifacts from artificial padding tokens. On character-level language modeling, DFT enables flexible heterogeneous designs (e.g., asymmetric QKV) and demonstrates superior convergence on variable-length batches compared to zero-padding. Analysis confirms that Projection-Padding effectively removes wasteful computations on padded positions. Although currently a proof-of-concept requiring optimization, DFT offers a promising direction for flexible, padding-free attention mechanisms.

Index Terms—Transformer, Semi-Tensor Product, Self-Attention, Variable-Length Sequences, Heterogeneous Data, Deep Learning, Padding-Free Methods.

I. INTRODUCTION

Since its inception, the transformer architecture has become a standard backbone for sequence modeling and large-scale pretraining [1]. Its self-attention mechanism is highly parallelizable and effective at capturing long-range dependencies, which has supported rapid progress in NLP—from bidirectional encoders such as bidirectional encoder representations from transformers (BERT) [2], to improved training recipes such as robustly optimized BERT approach (RoBERTa) [3], encoder–decoder pretraining such as text-to-text transfer transformer (T5) [4], and autoregressive generative models represented by GPT-2/GPT-3 [5], [6]. Beyond language, Transformers have also proven competitive in vision [7]–[9], speech recognition [10], [11], and biological sequence modeling [12]–[14].

The dominance of fixed-shape tensors. A practical driver of this success is that the standard Transformer is naturally expressed as compositions of matrix multiplications, linear projections, and elementwise nonlinearities. These operations map cleanly to GPU kernels, encouraging a data pipeline in which samples are packed into mini-batches, stacked into regular tensors, and processed by highly optimized linear-algebra routines. In effect, mainstream implementations assume that each batch can be made “rectangular”—both in sequence length and feature dimension—so that each layer executes in a uniform, vectorized manner.

Incompatibility with sequence heterogeneity. The rectangular-batch assumption, however, conflicts with the

heterogeneity of many real data sources. A first manifestation is variable-length sequence processing. In practice, padding and masking are widely used to unify batch shapes [1], [2]. While convenient, this choice introduces costs that become pronounced when length variance within a batch is large:

- 1) *Computational waste:* attention and feed-forward computation are performed on padded positions, and the overhead can dominate when the maximum length in a batch is much larger than the typical length [15]–[17].
- 2) *Statistical contamination:* padding tokens can perturb the statistics used by normalization layers and interact with training dynamics; the effect is often subtle, but it can be non-negligible in practice [18], [19].
- 3) *Optimization burden:* the model must learn to systematically suppress artificial tokens via masks and attention patterns, which may complicate optimization and slow convergence in long-context settings [20], [21].

These issues are not merely implementation details: they influence both what the model is asked to represent and where computation is spent. In particular, even with masking that removes attention to padded *keys*, the model still performs attention computation for padded *queries*; we quantify this effect later via post-hoc metrics.

Constraint of feature homogeneity. A second, less discussed limitation is the dimensional homogeneity required by dot-product attention. In the standard mechanism, the score term $QK^\top$ requires Query and Key representations to share the same feature dimension [1]. This constraint is often satisfied by design, but it also narrows the architectural search space. For example, in multimodal fusion, representations from different modalities (vision, text, audio) can have different intrinsic dimensionalities, and common practice projects them into a shared interaction space before fusion [22]–[28]. Such forced alignment introduces an additional design choice (the fusion dimension) and may act as an information bottleneck. Separately, from an efficiency viewpoint, the distinct roles of Q/K (similarity estimation) and V (content aggregation) motivate asymmetric designs, yet strict dimension matching makes these variants less direct to express in the conventional formulation [29], [30].

Generalizing algebraic primitives. Many Transformer improvements target the *pattern* or *approximation* of attention to reduce quadratic cost (e.g., sparsity, low-rank structure, kernel tricks), or accelerate exact attention through optimized kernels. These directions are valuable, but they typically retain the same algebraic primitives (dot products and dimension-matched MatMul) and still rely on batch rectangularization

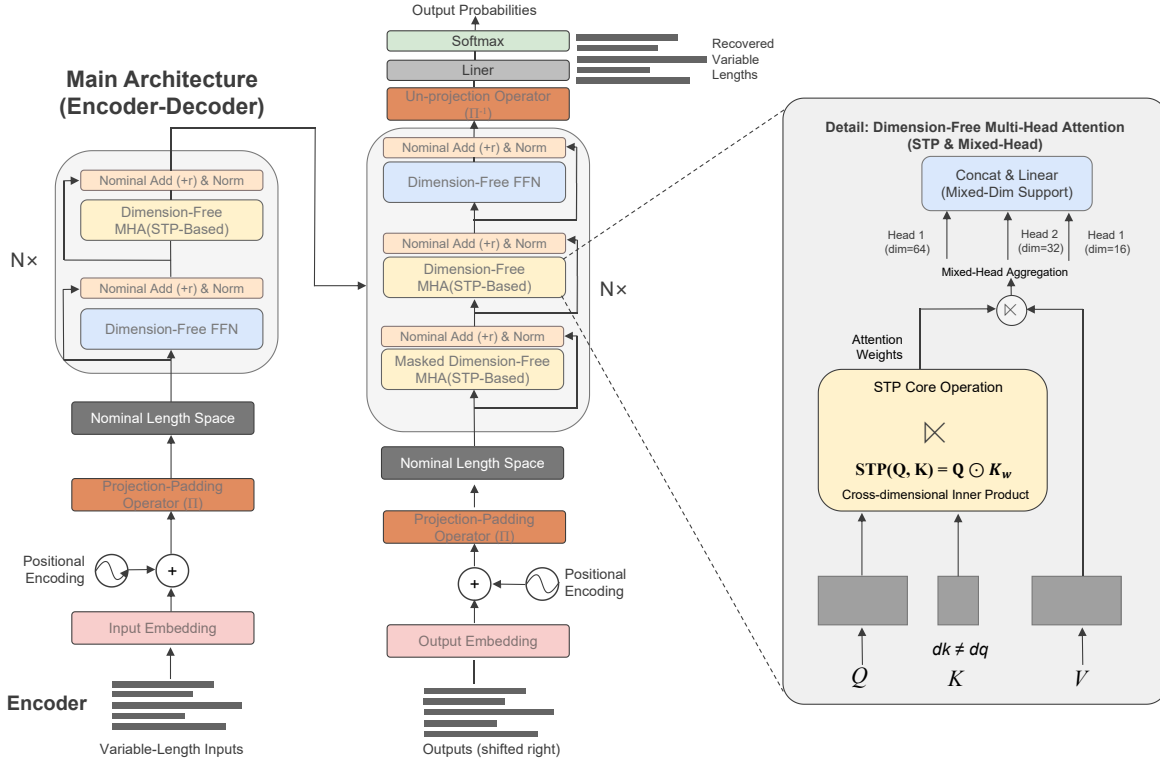


Fig. 1: The overall architecture of the proposed Dimension-Free Transformer (DFT). The main architecture (Left and Middle) adopts an Encoder-Decoder structure that integrates the Projection-Padding Operator (Π) and Un-projection Operator (Π^{-1}) to align variable-length inputs with the nominal length space. The detailed view (Right) illustrates the Dimension-Free Multi-Head Attention (STP-Based). By replacing the standard dot product with the Semi-Tensor Product (STP) operation (denoted as $\times$), the mechanism accommodates dimensionally heterogeneous inputs (i.e., $d_q \neq d_k$) and supports Mixed-Head Aggregation with varying head dimensions (e.g., 64, 32, 16).

via padding. The limitations above suggest a complementary approach: generalize the underlying linear-algebraic operations so that attention remains well-defined when either feature dimensions or sequence lengths are heterogeneous. STP provides a representative example of such a dimension-free algebraic primitive and has been surveyed with applications in several domains [31].

Proposed architecture: DFT. Motivated by this perspective, we introduce the DFT following the theoretical framework of Cheng [32]. The central idea is to integrate the STP [33]–[35], which generalizes conventional matrix multiplication to mismatched dimensions via Kronecker-based embeddings [36]. Beyond its theoretical role, STP has also been developed into a broader toolbox with applications in areas such as compressed sensing and related signal-processing settings [31], as well as stabilization problems in Boolean control networks [37]. Replacing dot-product-based components with STP/STA-based counterparts breaks the rigid dimensional coupling between Q , K , and V , enabling heterogeneous and asymmetric attention structures that are cumbersome to express under standard dimension-matching constraints.

Length unification via Projection-Padding. Building on the same framework, we further adopt a padding-free method termed Projection-Padding [32]. Instead of appending padding tokens, Projection-Padding applies a deterministic linear op-

erator that maps each sequence from its original length into a shared nominal-length space for batching, followed by an explicit un-projection that maps outputs back to the original length domain. Because no non-informative tokens are introduced, this mechanism removes padding-induced computation on artificial positions and mitigates padding-related statistical artifacts, while remaining compatible with token-level outputs.

Contributions. The main contributions of this paper are:

- 1) *Dimension-free formulation for Transformer blocks:* We incorporate STP/STA-based operators into self-attention, residual connections, and feed-forward transformations, yielding a Transformer architecture that does not require strict $Q/K/V$ dimension matching and supports principled modeling of heterogeneous representations.
- 2) *Projection-Padding for variable-length batches:* We replace zero-padding with deterministic projection/un-projection operators to unify sequence length without injecting artificial tokens, improving training efficiency and reducing padding-induced artifacts.
- 3) *Empirical validation and analysis:* Through controlled experiments, we verify equivalence to the standard Transformer in homogeneous settings, demonstrate new design space enabled by dimension freedom (e.g., mixed-head and asymmetric QKV designs), and show practical gains on variable-length modeling with quantitative and visual

mechanistic evidence.

The remainder of this paper is organized as follows. Section II reviews related work. Section III details STP preliminaries, the dimension-free attention mechanism, and Projection-Padding. Section IV presents and analyzes experimental results. Finally, Section V concludes and outlines future directions.

II. RELATED WORK

We review three lines of work most relevant to DFT:

- Efficiency-oriented Transformer variants;
- Techniques for variable-length sequence handling;
- Heterogeneous/multimodal representation learning.

A. Transformer Architectures and Their Variants

Since the original Transformer [1], self-attention has become a standard building block for sequence modeling, but its quadratic time and memory cost in sequence length ($O(N^2)$) remains a bottleneck for long contexts and dense tokenizations. A substantial body of work improves efficiency primarily by modifying the interaction pattern or approximating the softmax attention map. Representative directions include low-rank or projection-based approximations (e.g., Linformer [29]), kernelized/random-feature methods (e.g., Performer [38]), and sparse connectivity patterns (e.g., Longformer [15], BigBird [16], and earlier sparse designs [39]). Other approaches reduce cost via content-dependent grouping (Reformer [40]), linear-attention reinterpretations [30], and alternative approximations such as Nyström-based attention [41] or learned mixing mechanisms [42]. At the systems level, exact attention can be substantially accelerated using IO-aware kernels such as FlashAttention [43]. A broader overview is provided by Tay et al. [17].

In parallel, improvements in pretraining recipes and architectural refinements have driven practical performance. RoBERTa [3] highlights the impact of optimization and data scale beyond the original BERT setting [2]. T5 [4] provides a unified text-to-text formulation for diverse tasks. In vision, ViT [7] and data-efficient training refinements such as DeiT [8] demonstrate competitive performance with pure attention-based backbones, while hierarchical variants such as Swin [9] introduce stronger inductive bias for vision structure.

a) Difference from our work.: Most prior variants target *efficiency of attention computation* (via approximation, sparsity, kernelization, or kernel-level optimizations) while keeping the underlying dot-product algebra and the associated dimension-matching requirement intact. In contrast, DFT targets the attention primitive itself: by replacing dot products with STP-based constructions, attention can be defined without enforcing strict $Q/K/V$ feature homogeneity, and this flexibility can be further leveraged for heterogeneous head designs and padding-free length handling.

B. Methods for Handling Variable-Length Sequences

In practice, sequence lengths are rarely uniform. The dominant approach is still padding with masking in Transformer

implementations [1], [2]. While straightforward, padding introduces overhead that grows with length variance: computation is wasted on padded positions, and padded tokens may perturb the statistics used by normalization or interact with optimization dynamics [15], [19]. In long-context regimes, padding overhead compounds with quadratic attention, reinforcing the motivation for sparse or approximate long-range modeling

III. METHODOLOGY

This section presents the theoretical foundations and core mechanisms of the DFT. Our methodology follows Cheng’s dimension-free Transformer formulation [32], whose central idea is to replace conventional dimension-constrained linear operations in Transformers with STP/STA-based dimension-free counterparts from dimension-free matrix theory [33], [34]. We first introduce STP preliminaries, then describe dimension-free self-attention and the overall DFT block design, and finally detail Projection-Padding for variable-length sequences.

A. Preliminaries: Semi-Tensor Product

a) From dimension-matching to dimension-free linear algebra: Standard matrix multiplication requires matched inner dimensions. This constraint underlies the dimensional inflexibility in conventional Transformer formulations, where dot products and linear maps are defined over fixed Euclidean spaces. When dimensions differ, a common workaround is to insert explicit projections so that multiplication becomes well-defined. DFT adopts a different stance: it seeks a dimension-free primitive that can directly replace the key linear operations.

b) Definition and interpretation: To remove this restriction, we introduce STP [33]–[35], denoted by $\ltimes$. STP generalizes matrix multiplication by embedding operands into a common lifted space using Kronecker products [36]. Its formal definition is given below.

Definition 1 (STP). Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$ be arbitrary matrices. Let $t = \text{lcm}(n, p)$ be the least common multiple of n and p . The semi-tensor product of A and B is defined as:

$$A \ltimes B := (A \otimes I_{t/n})(B \otimes I_{t/p}), \quad (1)$$

where $\otimes$ denotes the Kronecker product and I_k is the $k \times k$ identity matrix.

c) Consistency with conventional multiplication: A key property is that STP reduces to the standard matrix product when $n = p$, ensuring backward compatibility with conventional architectures [34]. This recoverability is crucial for interpreting DFT as a strict generalization: in homogeneous settings, the model can collapse back to standard Transformer computations.

d) Computational note: The Kronecker-based lifting can introduce overhead in a direct implementation. In this paper, we treat the STP/STA framework as the theoretical backbone that exposes new architectural degrees of freedom [32]. Implementation-level acceleration and structured simplifications are engineering directions orthogonal to the core formulation.

B. Dimension-Free Self-Attention

a) *Limitation of scaled dot-product attention:* The standard scaled dot-product attention is $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$ [1]. Its core step $QK^\top$ requires Q and K to share the same feature dimension. Standard practice enforces this by choosing a fixed d_k and projecting all tokens into that space. Our objective is to remove this restriction so that attention is definable between representations of different intrinsic dimensions.

b) *Hypervector view of sequences:* We treat a sequence as a **hypervector**, i.e., an ordered set of vectors with potentially different dimensions, and build cross-dimensional similarity via STP-based constructions [32]. This perspective aligns with broader tensor/hyperstructure representations in multilinear algebra [44], [45]. Concretely, letting $X = \{x^1, \dots, x^T\}$ with $x^i \in \mathbb{R}^{d_i}$ allows both length and token feature dimension to vary without forcing a priori alignment.

c) *Definitions: cross-dimensional inner product and weighted score:* Let $q \in \mathbb{R}^m$ and $k \in \mathbb{R}^n$, and let $t = \text{lcm}(m, n)$. Define the lifted vectors $\tilde{q} = q \otimes \mathbf{1}_{t/m}$ and $\tilde{k} = k \otimes \mathbf{1}_{t/n}$. A cross-dimensional inner product can be defined as

$$\langle q, k \rangle := \frac{1}{t} \tilde{q}^\top \tilde{k}, \quad (2)$$

and a weighted variant (used for attention scoring) as

$$\langle q, k \rangle_W := \frac{1}{t} \tilde{q}^\top W \tilde{k}, \quad (3)$$

where $W \in \mathbb{R}^{t \times t}$ is a learnable weight matrix. This reduces to the standard dot product when $m = n$ (hence $t = m = n$).

d) *Step 1: score computation (dimension-free):* For each pair (i, j) , we compute a scalar similarity between $q^i \in \mathbb{R}^{d_q}$ and $k^j \in \mathbb{R}^{d_k}$ even when $d_q \neq d_k$, using the STP-based cross-dimensional inner product described in the source framework [32]. This yields a score matrix $S_{ij} = \langle q^i, k^j \rangle_W$, $S \in \mathbb{R}^{T \times T}$.

e) *Step 2: normalization (unchanged in form):* We apply softmax row-wise to obtain normalized attention weights $A = \text{softmax}(S)$. Softmax serves as a probabilistic normalization and does not require dimension matching of token features [46].

f) *Step 3: value aggregation via the $\diamond$ mapping:* In the standard Transformer, value aggregation is AV and requires V to be a regular matrix. In DFT, V is a hypervector, so aggregation is realized by a dimension-free linear mapping, denoted by $\diamond$, defined in the source framework as a composite operation of projection, matrix multiplication, and un-projection [32]. The complete mechanism is:

$$\text{Attention}(Q, K, V) = A \diamond V, \quad A = \text{softmax}(S), \quad (4)$$

where $\diamond$ denotes a projection-based transformation of a value hypervector:

$$A \diamond V := \left\{ \Pi_{d_i}^T \left(A \Pi_{d_i}^{d_i} v^i \right) \right\}_{i=1}^T. \quad (5)$$

g) *Reduction to the standard case:* When the sequence is homogeneous in feature dimension and the projections produce dimension-matched $Q/K/V$, the above construction collapses to the familiar scaled dot-product attention (up to the same normalization and nonlinear steps) [1], [32]. This property supports controlled ablations: replacing STP-based pieces by MatMul should recover standard behavior.

h) *Mixed-head attention as an architectural degree of freedom:* This freedom is especially meaningful in multi-head attention. Conventional multi-head attention partitions a single model dimension evenly across heads [1]. In DFT, heterogeneous head dimensions are allowed, enabling mixed-head configurations within one module [32]. Intuitively, heads may be assigned different representational budgets to capture information at different granularities without being forced into a uniform head dimension.

C. The Dimension-Free Transformer

a) *From a Transformer block to a DFT block:* A standard Transformer block consists of multi-head attention and a position-wise feed-forward network, each combined with residual connections and normalization [1]. Residual learning is a widely used stabilization strategy in deep networks [47]. In standard Transformers, residual addition assumes the input and output representations live in the same vector space; dimension changes typically require explicit projections.

b) *Core substitutions:* In our DFT block, the key components are systematically replaced by dimension-free counterparts derived from STP/STA theory [32]–[34]:

- **Dimension-Free Multi-Head Self-Attention:** STP-based score computation removes strict $Q/K/V$ dimension matching and supports heterogeneous head dimensions [32].
- **Dimension-Free Residual Connection:** standard vector addition is replaced by Nominal Addition (an STA-based construction) to support summation across different dimensions [32], [34]. This preserves the functional role of residual pathways while relaxing equal-dimensionality requirements.

$$x +_r y := \Pi_r^m x + \Pi_r^n y, \quad x \in \mathbb{R}^m, y \in \mathbb{R}^n. \quad (6)$$

- **Dimension-Free Feed-Forward Network:** linear transformations inside FFN are replaced by STP-based dimension-free linear maps, allowing FFN to accept and produce representations of arbitrary dimensions consistent with the overall framework [32].

c) *Nonlinearities remain dimension-preserving:* Nonlinear components such as ReLU and normalization do not change dimensionality and can be applied as usual [18], [48]. In practice, normalization behavior is sensitive to data statistics [19], which further motivates reducing padding-induced distribution artifacts at the input level.

D. Projection-Padding for Variable-Length Sequences

a) *Motivation: length-varying sequences without artificial tokens:* To address variable-length sequences, we adopt

Projection-Padding from the source framework [32]. The core idea is to define a nominal length T_{nominal} and map sequences of varying real lengths T_i into a shared representation space through deterministic linear projections. Unlike zero-padding, which appends non-informative vectors and then relies on masks, Projection-Padding uses a mathematically defined operator to transform the sequence into a common nominal-length space, together with a corresponding un-projection.

Definition 2 (Projection Matrix). For projecting a vector from an m -dimensional space to an n -dimensional space, the projection matrix $\Pi_n^m \in \mathbb{R}^{n \times m}$ is defined as [32]:

$$\Pi_n^m = \frac{n}{t} (I_n \otimes \mathbf{1}_{t/n}^T) (I_m \otimes \mathbf{1}_{t/m}), \quad (7)$$

where $t = \text{lcm}(m, n)$, I is the identity matrix, and $\mathbf{1}_k$ is an all-ones column vector of length k . Note that $\Pi_n^n = I_n$ and thus the operator reduces to the identity map when $m = n$.

b) Procedure: Given a batch $X = \{x^1, \dots, x^B\}$ with $x^i \in \mathbb{R}^{T_i \times d}$, Projection-Padding proceeds as:

- 1) Projection: $\tilde{x}^i = \Pi_{T_{\text{nominal}}}^{T_i} x^i$ to obtain a unified length T_{nominal} .
- 2) Core Processing: process $\tilde{X}$ with a standard Transformer or DFT block.
- 3) Un-projection: map back via $\Pi_{T_i}^{T_{\text{nominal}}}$ to recover outputs in the original length space.

c) Interpretation and practical remarks: Compared with zero-padding, Projection-Padding performs length unification via deterministic linear operators rather than by injecting artificial tokens, thereby avoiding padding-induced noise and attention waste at the source. The nominal length T_{nominal} can be chosen based on application constraints (e.g., a fixed compute budget), and the explicit un-projection keeps the method compatible with token-level outputs, unlike pooling-based compression [4]. From a modeling standpoint, the mechanism aims to keep the data stream “clean” so that attention and normalization operate on transformed informative content rather than on a mixture of content and injected zeros.

IV. EXPERIMENTS AND ANALYSIS

This section evaluates the proposed Dimension-Free Transformer (DFT). We conduct a set of controlled experiments to (i) verify that STP-based attention recovers standard behavior in homogeneous settings, (ii) examine whether dimension freedom expands the design space in a useful way, and (iii) assess its impact on variable-length sequence modeling. The evaluation is organized around four research questions:

- RQ1 (Foundational equivalence). On fixed-length, homogeneous inputs, does STP-based attention match a conventional Transformer that uses standard matrix multiplication?
- RQ2 (Architectural flexibility). Does the dimension-free formulation enable heterogeneous head dimensions (e.g., mixed-head attention), and does this flexibility translate into measurable gains under a fixed parameter budget?

- RQ3 (Variable-length modeling). On variable-length batches, does Projection-Padding improve optimization behavior (e.g., convergence speed/stability) and final validation loss compared with the standard Zero-Padding baseline?
- RQ4 (Mechanism). Can we explain the observed differences by analyzing attention behavior, using both visualization and quantitative post-hoc metrics, and verify that Projection-Padding reduces padding-induced computation on non-informative positions?

We first describe the datasets, training setup, and model configurations in Section IV-A. We then report the fixed-length equivalence study (RQ1) in Section IV-B, evaluate mixed-head configurations (RQ2) in Section IV-C, analyze asymmetric QKV designs (RQ2) in Section IV-D, and finally compare padding strategies together with mechanistic analysis (RQ3–RQ4) in Section IV-E.

A. Experimental Setup

This subsection specifies the datasets, preprocessing, model configurations, and training protocol used in our experiments. **Datasets and Pre-processing.** Our primary experimental corpus is the publicly available `shakespeare_char` dataset, which comprises the complete works of William Shakespeare. The raw text was pre-processed into a vocabulary of 65 unique characters. To simulate real-world length heterogeneity, we construct a variable-length version of the dataset (`shakespeare_char_hetero`) by randomly sampling contiguous text segments with true length $L \in [16, 256]$ (unless otherwise stated). The resulting token streams are stored as binary files (e.g., `train.bin` and `val.bin`) and are used for both training and evaluation of padding strategies. **Model Configurations.** We use two controlled model settings to keep comparisons fair while enabling efficient ablations.

Fixed-length equivalence setting (RQ1). For the foundational equivalence test, we use a standard small Transformer configuration with `block_size=256` and identical hyperparameters across the baseline and DFT models; the only difference is the attention implementation (MatMul vs. STP-based).

Design-space and variable-length setting (RQ2–RQ4). For mixed-head, asymmetric QKV, and padding-strategy experiments, we adopt a compact configuration (approximately 0.829M parameters, `block_size=256`) to allow repeated controlled runs. In these experiments, compared models share the same optimizer and training schedule, and differ only in the architectural choice being tested (e.g., head dimension allocation, $Q/K/V$ dimensions, or padding strategy). We set `bias=False` and `dropout=0.0` in all runs to reduce confounding factors.

Training and Evaluation Details. Models were trained using the AdamW optimizer with hyperparameters set to $\beta_1 = 0.9$, $\beta_2 = 0.95$, and a weight decay of 0.1. The learning rate was managed by a cosine decay schedule, starting from an initial rate of 6×10^{-4} and decaying to a minimum of 6×10^{-5} over 600,000 training iterations, preceded by a linear warmup phase of 2000 iterations. We used a per-device batch size of

12 and applied gradient accumulation over 40 steps, achieving an effective batch size of 480 samples. To enhance training stability, gradient clipping was applied with a maximum norm threshold of 1.0. Model performance was evaluated every 2000 iterations by computing the average loss over 200 batches from both the training and validation sets to obtain stable and reliable performance metrics.

Implementation Environment. All experiments were conducted on a local workstation equipped with a single NVIDIA GeForce RTX 4060 Ti GPU (8GB GDDR6 VRAM). The implementation was based on PyTorch (v2.1.0) with CUDA (v12.1). We used `bfloat16` mixed-precision training and enabled `torch.compile` for JIT compilation.

B. Foundational Validation: Equivalence in Standard Scenarios

Motivation and hypothesis. As a prerequisite, we test whether STP-based attention behaves like standard dot-product attention in the homogeneous, fixed-length setting. Specifically, we ask whether a DFT model matches a conventional Transformer when the input tensors have fixed length and matched feature dimensions. Under this setting, our hypothesis is that replacing the attention implementation (MatMul vs. STP-based computation) should not materially change optimization dynamics or final validation loss.

Our central hypothesis is that if our STP-based attention mechanism is a correct and valid generalization of standard dot-product attention, then its computational results and learning dynamics should exhibit no significant difference from the standard model under the special case of uniform input tensor dimensions. Confirming this hypothesis is critical. It not only verifies the correctness of our core algorithm but also demonstrates that the DFT is not an isolated architecture but rather a logical extension of existing ones. This establishes the credibility required for subsequent experiments that explore its unique advantages in handling variable-dimension and variable-length sequences.

Experimental Design. To rigorously test this hypothesis, we designed a tightly controlled comparative experiment.

- **Data Configuration:** We utilized the `shakespeare_char` dataset, as described in Section IV-A, but employed a standard preprocessing pipeline that formats all training and validation samples into sequences of a fixed length (`block_size = 256`). This ensures the input data conforms perfectly to the operational requirements of a standard Transformer.
- **Model Comparison:** We conducted a direct comparison between our proposed DFT model and a standard Transformer baseline. To ensure a fair comparison, both models shared an identical hyperparameter configuration (e.g., $n_{\text{layer}} = 6$, $n_{\text{head}} = 6$, $n_{\text{embd}} = 384$) and were trained using the exact same optimizer settings, learning rate schedule, and training procedure. The sole distinction between the two models was the underlying implementation of the attention sub-layer: one using conventional Batched Matrix Multiplication, and the other using our proposed STP-based implementation.

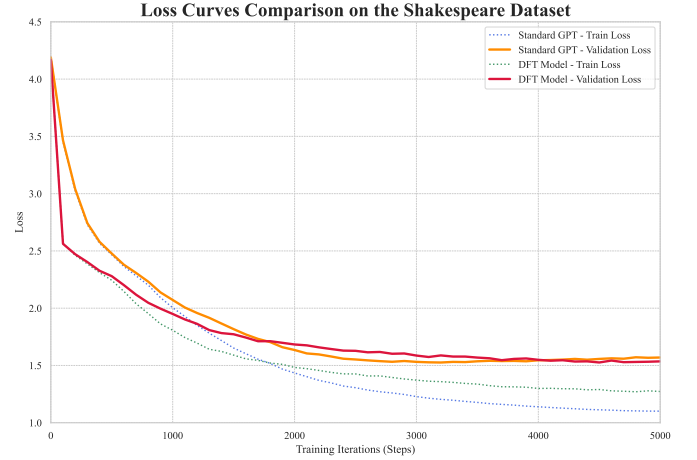


Fig. 2: Validation loss on the fixed-length setting for a standard Transformer and DFT. The two curves closely overlap throughout training, indicating that STP-based attention recovers the behavior of conventional attention under homogeneous inputs.

Results and Analysis. The quantitative results of this experiment are presented in Figure 2 shows the validation loss trajectories of the two models. The curves closely track each other over training and converge to nearly the same final level (e.g., 1.452 vs. 1.455). This result supports the intended reduction property of the formulation: in the homogeneous case, STP-based attention does not degrade performance relative to the standard implementation.

This outcome provides strong empirical support for our initial hypothesis. It demonstrates that although the DFT is engineered to handle more complex, heterogeneous data, its performance is not compromised in the simplified, homogeneous data scenario. This empirically confirms that our STP-based attention mechanism is a correct and lossless generalization of the standard attention paradigm. With this foundational validation complete, we can now proceed with confidence to investigate the unique capabilities and advantages that the DFT framework offers in scenarios that are intractable for traditional architectures.

C. Empirical Validation of Dimension-Free Flexibility via Mixed-Head Attention

a) Motivation.: In standard multi-head attention, all heads typically share the same head dimension, mainly for implementation convenience under dot-product attention. In DFT, STP-based scoring relaxes this requirement and allows different heads to use different dimensions within the same layer. We test whether this additional degree of freedom can be exploited under a fixed parameter budget.

b) Experimental Design.: We compare two DFT configurations on `shakespeare_char_hetero`. Both models share the same training setup and total embedding size ($n_{\text{embd}} = 128$); they differ only in how the 4 head dimensions are allocated. This isolates the effect of heterogeneous head dimensions.

The two configurations were designed to be identical in all aspects—including total embedding dimension ($n_{\text{embd}} =$

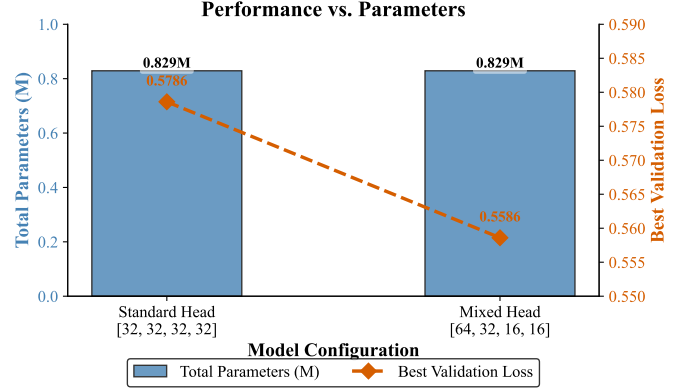
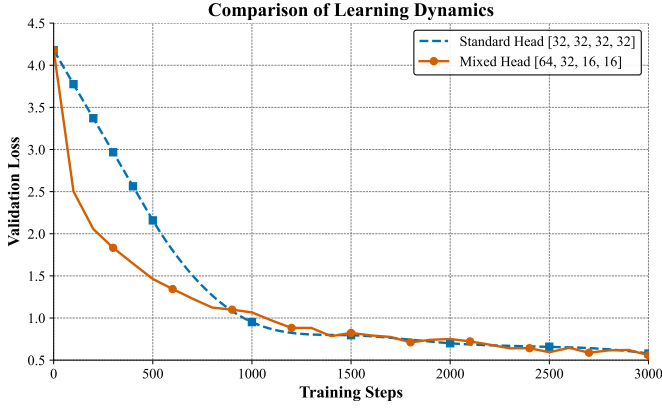


Fig. 3: Mixed-head attention results. **(Left)** Validation loss over 3000 steps for the mixed-head and uniform-head configurations. **(Right)** Best validation loss under the same parameter budget.

TABLE I: Performance and parameter comparison of DFT configurations. The Mixed-Head model achieves a lower validation loss with an identical parameter budget.

Model Configuration	Total Parameters (M)	Best Validation Loss ↓
Standard-Head DFT	0.829	0.5786
Mixed-Head DFT	0.829	0.5586

128), number of layers, and training hyperparameters—differing only in the internal allocation of dimensions across their four attention heads:

- **Standard-Head DFT (Baseline):** A control model with 4 attention heads of uniform dimension, specifically $[32, 32, 32, 32]$, representing the conventional homogeneous design.
- **Mixed-Head DFT (Ours):** An experimental model leveraging dimensional freedom, configured with 4 heads of heterogeneous dimensions: $[64, 32, 16, 16]$. This structure hypothetically allows a 64-dimension head to capture broader context, while the 16-dimension heads focus on fine-grained patterns.

Both models were trained for 3000 iterations. We compare them based on their parameter count, final validation loss, and learning dynamics.

c) Results and Analysis.: The quantitative results of our experiment are summarized in Table I, with the learning dynamics and final performance visualized in Figure 3.

The results are summarized in Table I and Figure 3. Both configurations have the same parameter count (0.829M), so the performance difference is not due to model scale. The mixed-head configuration achieves a lower best validation loss (0.5586 vs. 0.5786), corresponding to a 3.5% relative reduction.

A consistent gap is also visible in the loss trajectories after the initial optimization phase (approximately the first 200 steps). These observations suggest that allocating head dimensions non-uniformly can be beneficial under a fixed budget, and DFT provides a direct way to express such designs without introducing additional projection constraints.

D. Parameter Efficiency Analysis: Asymmetric Attention Architectures

Motivation and question. In standard self-attention, the Q , K , and V projections are typically configured with matched dimensions, partly because dot-product scoring requires $d_q = d_k$. However, Q/K mainly serve to form attention weights (similarity estimation), while V carries the content being aggregated. This asymmetry motivates a simple question: under a fixed training setup, can we reduce the dimensionality of K (and optionally reallocate the saved parameters to V) without degrading validation performance?

Experimental Design. To systematically investigate this question, we designed a controlled experiment comparing three model configurations, all built upon our strictly theoretical DFT framework. Each model was trained for 3000 iterations on the heterogeneous Shakespeare dataset.

- **Symmetric DFT (Baseline):** This model serves as our baseline, adhering to a conventional symmetric design where the dimensions for Q , K , and V are identical ($d_q = d_k = d_v = 128$).
- **K-Compressed DFT:** This configuration tests the hypothesis of Key matrix redundancy. We reduce the dimension of K by half ($d_k = 64$) while keeping Q and V dimensions unchanged ($d_q = 128, d_v = 128$), directly reducing the parameter count of the Key projection matrix.
- **Reallocated DFT:** This model explores a more aggressive strategy of parameter reallocation. We compress the K dimension ($d_k = 64$) and reinvest the parameter budget into the V dimension, enhancing its representational capacity ($d_v = 192$) while maintaining the Q dimension ($d_q = 128$).

All other architectural hyperparameters, such as the number of layers (4) and heads (4), and the training configuration were held constant across all experiments to ensure a fair comparison.

Results and Analysis. Table II and Figure 4 summarize the outcomes. Reducing the Key dimension improves validation loss while lowering parameters: the K-Compressed configuration uses 7.00% fewer parameters and achieves a 10.92% lower best validation loss than the symmetric baseline.

TABLE II: Quantitative results of the parameter efficiency analysis. The K-Compressed model achieves superior performance with fewer parameters, while the Reallocated model demonstrates a substantial performance gain with a negligible parameter increase.

Model Configuration	Q-K-V Dimensions	Total Params (M)	Param Change (%)	Best Val. Loss	Loss Change (%)
Symmetric DFT (Baseline)	128-128-128	0.829	-	0.5797	-
K-Compressed DFT	128-64-128	0.771	-7.00%	0.5164	-10.92%
Reallocated DFT	128-64-192	0.837	+0.97%	0.4436	-23.48%

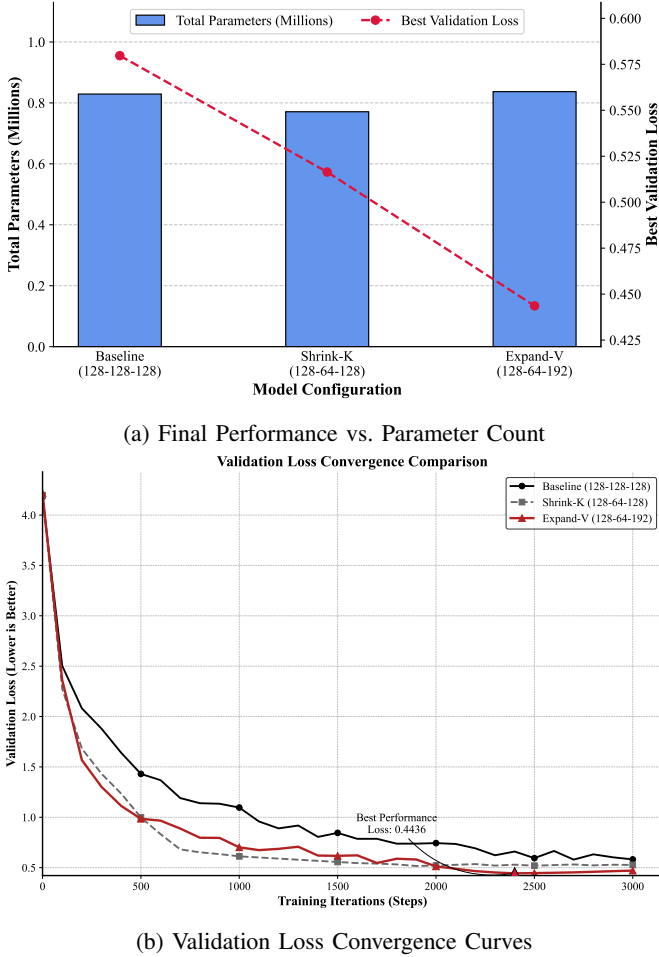


Fig. 4: Parameter efficiency and learning dynamics of asymmetric DFT models. (a) A comparison of total model parameters (blue bars, left axis) and final best validation loss (red markers, right axis). (b) The convergence behavior of the three models over 3000 training iterations. The Reallocated DFT model not only achieves the best final performance but also demonstrates a consistently lower loss throughout the training process.

Reallocating parameters from K to V yields the best result. The Reallocated DFT has a parameter count close to the baseline (+0.97%) but achieves a substantially lower validation loss (0.4436, a 23.48% reduction). The convergence curves in Figure 4(b) show that this improvement is not only a final-point effect: the Reallocated model maintains a lower loss for most of training.

E. Projection-Padding for Variable-Length Sequences

1) *The Challenge of Variable-Length Data:* In many applications, inputs have variable lengths, while GPU-efficient training typically assumes regular tensors with a common shape. The prevailing workaround is zero-padding, which pads shorter sequences to a fixed length within a batch and uses masks to block attention to padded *keys*. Although simple, this approach performs substantial computation on padded positions and may interact with training dynamics through auxiliary activations that are unrelated to the underlying data. This section evaluates whether Projection-Padding provides a practical alternative in the same setting.

2) *Performance Comparison: Projection-Padding vs. Zero-Padding: Experimental Design.* We perform a controlled comparison on `shakespeare_char_hetero`, where each sample has a true length $L \in [16, 256]$. The two conditions differ only in the length-unification strategy:

- **Zero-Padding (Baseline):** Each batch is padded to a fixed context length (`block_size=256`). An attention mask is applied to prevent valid tokens from attending to padded *keys*.
- **Projection-Padding (Ours):** Each sequence is projected to a nominal length (`nominal_dim=256`) via the Projection-Padding operator and then processed in the nominal space, followed by an un-projection back to the original length.

Both models use the same architecture (0.829M parameters), optimization settings, and evaluation protocol. We report validation loss and convergence behavior.

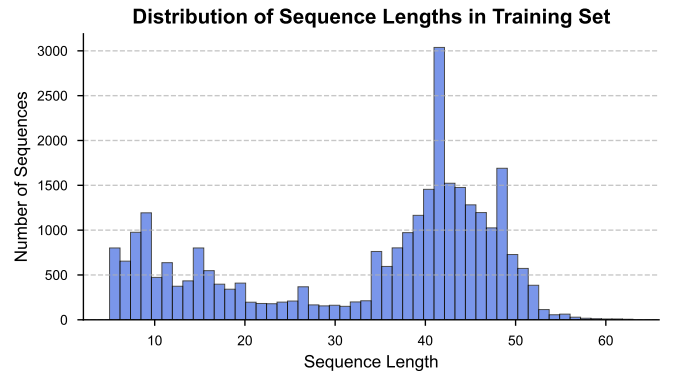


Fig. 5: Empirical distribution of true sequence lengths L in the heterogeneous Shakespeare validation split. The wide length variability motivates padding-aware designs.

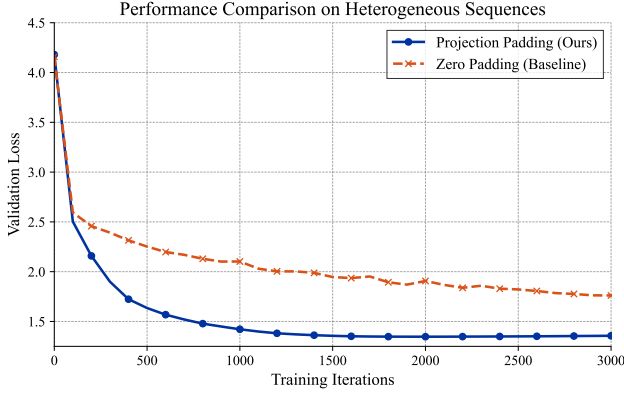


Fig. 6: Validation loss curves for models trained with Projection-Padding versus Zero-Padding.

TABLE III: Quantitative comparison of key performance metrics for the Zero-Padding (Baseline) and Projection-Padding (Ours) methods.

Metric	Baseline	Ours
Parameters (M)	0.829	0.829
Best validation loss	1.7610	0.5271
Iterations to val. loss < 1.9	≈ 1850	≈ 300
Val. loss at 3000 iterations	1.7610	0.5311

Results and Analysis. Figure 6 and Table III summarize the comparison. Projection-Padding reaches a validation loss below 1.9 much earlier (approximately 300 iterations) than Zero-Padding (approximately 1850 iterations), corresponding to about a $6\times$ reduction in iterations to this threshold under the same training configuration. Projection-Padding also attains a substantially lower best validation loss (0.5271 vs. 1.7610), and the gap remains throughout training.

A plausible explanation involves both computational and optimization factors. Mechanistically, Projection-Padding avoids executing attention and feed-forward computation on padded positions, whereas Zero-Padding must still process padded *query* rows even when padded *keys* are masked. Beyond direct computation, this result highlights the *optimization burden* imposed by padding: the baseline model must implicitly learn to suppress the influence of artificial tokens—a non-trivial task that consumes representational capacity and gradient updates. Furthermore, Projection-Padding mitigates *statistical contamination* in normalization layers. Standard padding injects non-informative zeros that can skew the feature statistics (e.g., mean and variance) used by LayerNorm. By mapping variable inputs to a nominal space without such artifacts, our method ensures that the network operates solely on semantically meaningful content.

3) Mechanism Analysis: Attention Visualization and Quantification:

Length Heterogeneity in the Evaluation Set. To contextualize the impact of padding, we first visualize the empirical length distribution of the heterogeneous validation split. As shown in Figure 5, the dataset contains a wide range of sequence lengths, implying that any fixed-length padding scheme will incur substantial overhead for shorter sequences.

Qualitative Evidence: Attention Heatmaps. To provide an intuitive view of padding effects, we extract attention weights from the final Transformer layer using the same input sequence with true length $L = 45$. The resulting attention patterns are shown in Figure 7.

Clarification: masking removes padded keys, not padded queries. In Zero-Padding, an attention mask can prevent valid queries from assigning probability mass to padded *key* positions. However, the model still computes attention scores and attention outputs for padded *query* positions (i.e., the extra $(T - L)$ rows induced by padding to length T). These rows are not tied to meaningful tokens, but they still trigger the same attention/FFN kernels and produce intermediate activations. Projection-Padding avoids this class of padded-query computation by construction, because the computation is performed in a representation that is matched to the effective content rather than extended by explicit padding tokens.

Quantitative Evidence: padded-query fraction and entropy ratio. To quantify the mechanism beyond a single example, we analyze 300 randomly sampled validation sequences. For each sequence with true length L and padded length $T = 256$, we report:

$$\text{PQR}(L) = \frac{T - L}{T}, \quad (8)$$

the *padded-query row fraction*, which measures the fraction of query rows that are introduced purely by padding (and therefore necessarily computed under a standard $T \times T$ attention implementation), and

$$\text{EntropyRatio}(L) = \frac{\frac{1}{T-L} \sum_{i=L+1}^T \left[-\sum_{j=1}^T A_{ij} \log(A_{ij} + \epsilon) \right]}{\frac{1}{L} \sum_{i=1}^L \left[-\sum_{j=1}^T A_{ij} \log(A_{ij} + \epsilon) \right]}, \quad (9)$$

which compares the average attention entropy of padded-query rows to that of valid-query rows (we use $\epsilon = 10^{-12}$ for numerical stability).

Figures 8–9 and Table IV summarize the results. As expected, $\text{PQR}(L)$ is largest for short sequences and decreases as L approaches T . In addition, padded-query rows exhibit consistently higher attention entropy on average, indicating more diffuse and less structured attention patterns on computations induced by padding.

Takeaway. The heatmaps in Figure 7 provide an intuitive, qualitative illustration; meanwhile, PQR and EntropyRatio offer quantitative evidence linked to sequence length and attention patterns. Through these experimental demonstrations, taken together, they support the view that a key cost of Zero-Padding arises from computations on padded query rows, which are not removed by masking. In contrast, Projection-Padding avoids explicit padding tokens and therefore reduces this kind of padding-induced extra computation. This is also consistent with the faster convergence and the lower final validation loss observed in Section IV-E2. It reflects that Zero-Padding entails some waste to a certain extent, and that our Projection-Padding can effectively avoid attention-related losses. These results highlight practical benefits from eliminating unnecessary padded-query computation.

TABLE IV: Quantitative post-hoc analysis on the heterogeneous validation set ($T = 256$). **PQR** is the padded-query row fraction $(T - L)/T$, and EntropyRatio compares the attention entropy of padded-query rows to valid-query rows. **Theo. Compute Waste** is computed from length only (e.g., $1 - (L/T)^2$ for full $T \times T$ score computation).

Length Bin	Count	L_{mean}	PQR (mean)	PQR (std)	EntropyRatio (mean)	EntropyRatio (std)	Theo. Compute Waste
[16, 32]	20	22.35	0.912695	0.017306	1.680357	0.057019	0.992078
[32, 64]	36	48.1389	0.811957	0.033262	1.729110	0.046123	0.963534
[64, 96]	35	80.9143	0.683929	0.034968	1.693674	0.044922	0.898876
[96, 128]	42	111.119	0.565941	0.034774	1.665653	0.063214	0.810384
[128, 160]	39	143.4872	0.439503	0.038384	1.641218	0.060862	0.684370
[160, 192]	47	176.4043	0.310921	0.032965	1.619545	0.066202	0.524083
[192, 224]	41	208.2195	0.186643	0.039286	1.570011	0.056475	0.336906
[224, 256]	40	241.8	0.055469	0.033876	1.541650	0.138641	0.106713

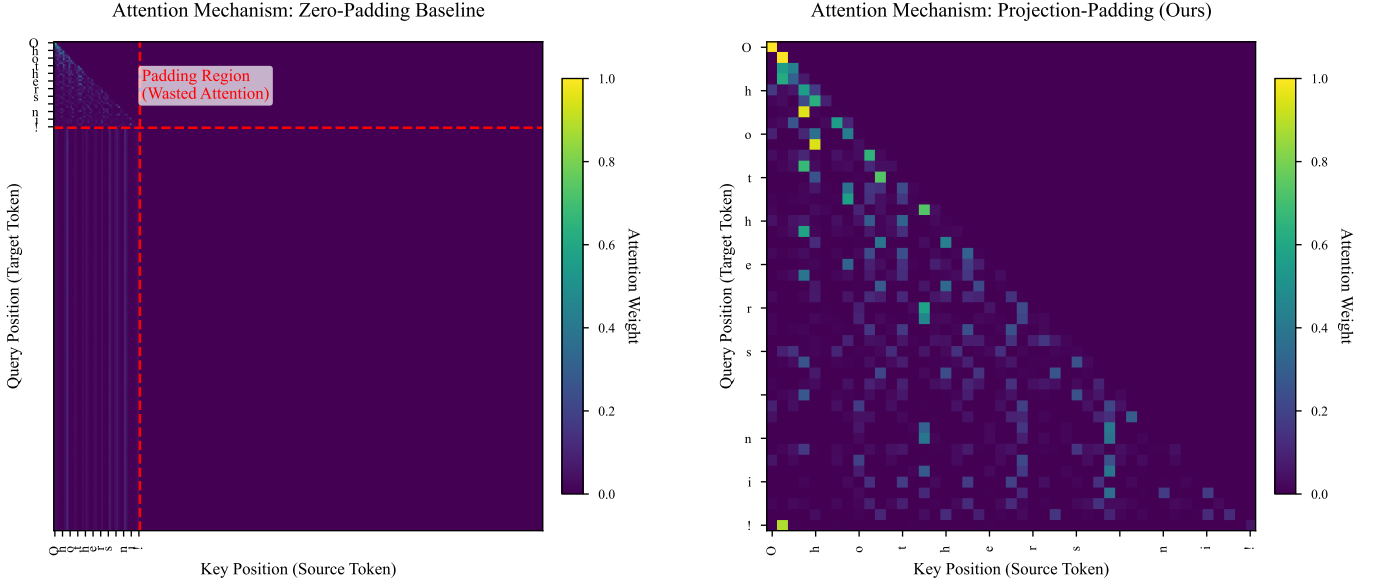


Fig. 7: Attention visualization for a sequence of 45 tokens. (Left) Zero-Padding yields a fixed $T \times T$ matrix ($T = 256$); positions beyond the dashed boundary correspond to padded *query* rows. (Right) Projection-Padding yields a compact $L \times L$ matrix aligned with the true length.

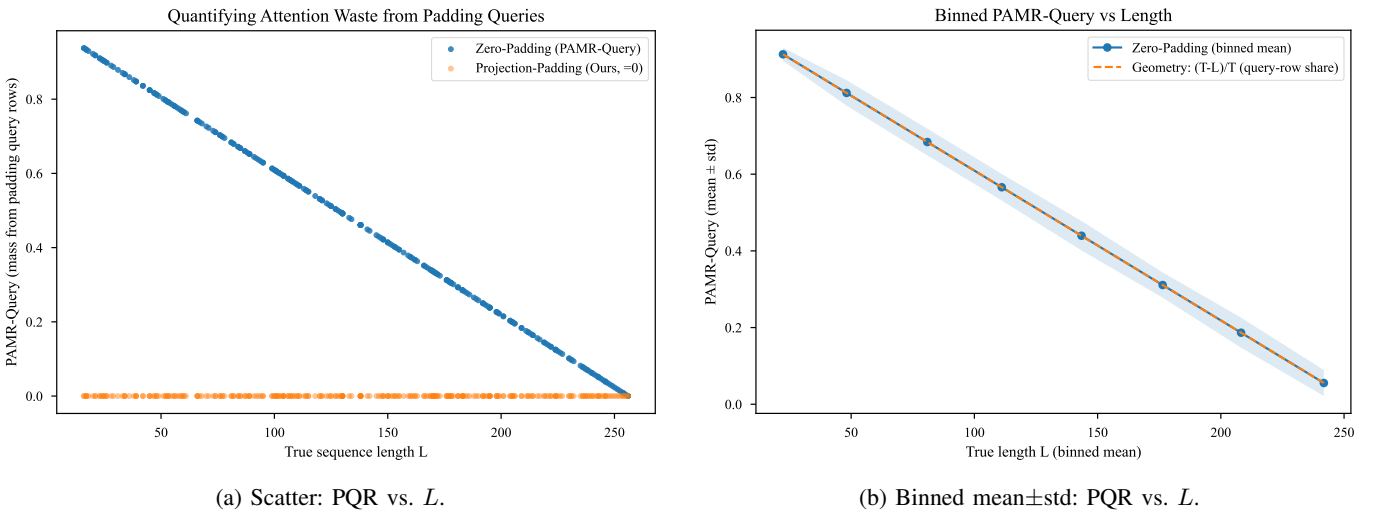


Fig. 8: Quantifying padding-induced compute on padded queries in Zero-Padding using PQR. The padded-query fraction is highest for short sequences and decreases as L approaches T . Projection-Padding does not create explicit padded query rows.

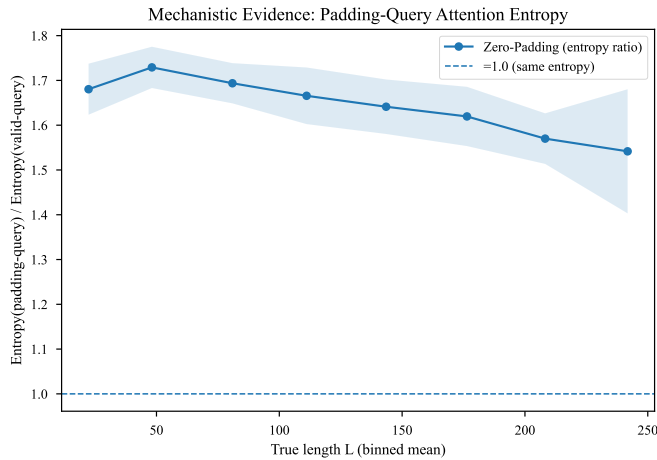


Fig. 9: Binned mean $\pm$ std of EntropyRatio. Padded query rows consistently exhibit higher attention entropy than valid query rows, suggesting more diffuse (less structured) attention behavior on padding-induced computations.

V. CONCLUSION

The standard Transformer is highly effective on regular, fixed-shape tensors, but this assumption can be restrictive for heterogeneous data with varying feature dimensions and sequence lengths. In this work, we proposed the DFT, which replaces the dot-product-based components in self-attention with STP-based constructions, thereby relaxing the strict dimension-matching requirement among Q , K , and V . Building on the same formulation, we introduce Projection-Padding, a projection/un-projection strategy that enables variable-length batching without injecting explicit padding tokens.

Our experiments provide evidence in three directions. First, in the homogeneous fixed-length setting, DFT exhibits convergence behavior and validation performance comparable to a standard Transformer baseline, supporting the intended reduction to the conventional case. Second, the dimension-free formulation makes it straightforward to express architectural variants such as mixed-head configurations, and we observe consistent gains under a fixed parameter budget in our controlled setup. Third, on variable-length sequence modeling, Projection-Padding leads to faster convergence and lower validation loss than Zero-Padding under the same training protocol. Mechanistic analyses based on attention visualizations and post-hoc metrics further suggest that these gains are closely related to reducing computation associated with padding-induced query positions.

A current limitation is that our proof-of-concept implementation incurs non-trivial overhead. We attribute this mainly to an implementation mismatch: DFT naturally operates on heterogeneous structures (e.g., lists of tensors with different shapes), whereas GPU kernels are typically optimized for large, regular tensors. A primary direction for future work is therefore performance engineering, including custom CUDA kernels and/or compiler-assisted implementations (e.g., Triton [49]). With improved efficiency, we plan to evaluate DFT on broader heterogeneous domains such as multimodal

learning and graph-structured data, and to further study how different dimensional allocations affect performance and efficiency.

Overall, DFT offers a principled extension of Transformer computation that supports heterogeneous dimensions and lengths within a unified framework, and it opens an additional axis for architectural design beyond the conventional homogeneous-tensor paradigm.

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